
2-Mark Questions & Answers

Q1. Define Data with an example.

Data refers to raw, unprocessed facts and figures that have no meaning on their own.

Example: 21, Suraj, Mumbai are individual pieces of data. They have no context until organized.

Q2. Define Database with an example.

A **Database** is an organized, structured collection of related data stored so it can be easily accessed, managed, and updated.

Example: A college database stores student records — Roll No, Name, Branch, Marks — all inter-related and organized.

Q3. Define DBMS.

A **Database Management System (DBMS)** is system software that enables users to define, create, maintain, and control access to a database. It acts as an interface between the user and the stored data.

Example: MySQL, Oracle, PostgreSQL.

Q4. What is the aim of a DBMS?

The primary aim of a DBMS is to provide a convenient and efficient environment to **store, retrieve, and manage data** while ensuring **data integrity, security, consistency, and reduced redundancy**.

Q5. Give any four examples of DBMS software.

1. **MySQL** — Open-source relational DBMS.
 2. **Oracle Database** — Enterprise-level RDBMS.
 3. **PostgreSQL** — Advanced open-source RDBMS.
 4. **Microsoft SQL Server** — RDBMS by Microsoft.
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Q6. What is a Query Processor?

The **Query Processor** is a DBMS component that interprets and translates high-level user queries (written in SQL) into low-level instructions that the storage manager can execute. It includes the DDL interpreter, DML compiler, and query evaluation engine.

Q7. What is a Storage Manager?

The **Storage Manager** is a DBMS component responsible for interfacing with the OS file manager to store, retrieve, and update data on physical disk storage. It manages the **authorization manager, integrity manager, transaction manager, file manager, and buffer manager**.

Q8. Define Data Independence.

Data Independence is the ability to modify the schema at one level of the database system without affecting the schema at the next higher level. It ensures that changes in storage or logical structure do not impact the application programs.

Q9. What is Logical Data Independence?

Logical Data Independence is the ability to change the **conceptual/logical schema** (e.g., adding a new attribute or table) without requiring changes to the **external/view-level schema** or the application programs that use it.

Q10. What is Physical Data Independence?

Physical Data Independence is the ability to change the **internal/physical schema** (e.g., changing the storage structure, file organization, or indexing method) without requiring changes to the **conceptual/logical schema**.

Q11. What are Naive Users?

Naive users are unsophisticated end-users who interact with the database through pre-built application interfaces (forms/GUIs) without knowing the underlying DBMS. They do not write SQL queries.

Example: A bank teller using a banking application to check customer balance.

Q12. Who are Sophisticated Users?

Sophisticated users are engineers, scientists, or business analysts who interact directly with the DBMS by writing SQL queries or using tools like report generators. They understand the database structure well.

Example: A data analyst writing complex SQL queries to generate business reports.

Q13. Define Database Administrator (DBA).

A **Database Administrator (DBA)** is a person who has central control over the database system. The DBA is responsible for defining the schema, granting access, monitoring performance, handling backups, and ensuring the database runs efficiently and securely.

Q14. What is the View Level in data abstraction?

The **View Level** (External Level) is the highest level of data abstraction. It describes only a **part of the entire database** that is relevant to a particular user or group of users. Different users see different views of the same data.

Q15. What is the Physical Level in data abstraction?

The **Physical Level** (Internal Level) is the lowest level of data abstraction. It describes **how data is actually stored** on the physical storage medium — including file organization, indexing structures, and access paths.

Q16. What is 1-tier architecture?

In **1-tier architecture**, the database, the application logic, and the user interface all reside on the **same single machine**. There is no network involved. It is used for local/personal applications.

Example: MS Access running on a standalone PC.

Q17. Name the two types of Data Independence.

1. **Logical Data Independence** — Changing the logical schema without affecting the view level.
 2. **Physical Data Independence** — Changing the physical schema without affecting the logical level.
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Q18. What is a Data Dictionary?

A **Data Dictionary** (also called a System Catalog) is a repository that stores **metadata** — information about the structure of the database such as table names, column names, data types, constraints, and relationships. It is maintained by the DBMS automatically.

Q19. What is an Index in a database?

An **Index** is a data structure (similar to a book's index) maintained by the Storage Manager to enable **fast retrieval** of records from a database table based on specific column values, avoiding a full table scan.

Q20. Define Concurrency Control (in one line).

Concurrency Control is the mechanism used by a DBMS to manage simultaneous access to the database by multiple users to ensure data **consistency and isolation** without conflicts.

Q21. Define Entity and Entity Set with an example.

- **Entity:** A real-world object or concept that is distinguishable from other objects based on its attributes. Example: A specific student "Suraj" with Roll No 101.
 - **Entity Set:** A collection of similar entities of the same type that share the same properties or attributes. Example: The set of all students in a university (STUDENT).
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Q22. Define Relationship and Relationship Set with an example.

- **Relationship:** An association among two or more entities. Example: Student "Suraj" *enrolls in* Course "DBMS".
 - **Relationship Set:** A set of relationships of the same type. Example: The ENROLLS_IN relationship set linking entities from the STUDENT entity set to the COURSE entity set.
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Q23. What are the three basic concepts of the ER model?

1. **Entity Sets:** Real-world objects/concepts (e.g., STUDENT, COURSE).
 2. **Attributes:** Properties or characteristics describing an entity (e.g., Roll_No, Name).
 3. **Relationship Sets:** Associations connecting two or more entity sets (e.g., ENROLLS_IN).
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Q24. Define Attribute in the ER model. What is a Domain of an attribute?

- **Attribute:** A property or characteristic that describes an entity or relationship set (e.g., Age, Salary).
 - **Domain:** The set of all permitted values for a given attribute. Example: The domain of Age is positive integers between 1 and 120.
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Q25. Define Degree of a relationship (with binary/ternary examples).

The **Degree of a relationship** refers to the number of entity sets participating in a single relationship set.

* **Binary Relationship (Degree 2):** Involves 2 entity sets. Example: STUDENT *enrolled in* COURSE.

* **Ternary Relationship (Degree 3):** Involves 3 entity sets. Example: SUPPLIER *supplies* PART to PROJECT.

Q26. What is meant by "Role" in a relationship?

A **Role** is the explicit function or role that an entity set plays in a relationship. Roles are typically implicit, but they are explicitly labeled on ER diagram lines when the same entity set participates more than once in a recursive relationship. Example: In a PREREQUISITE relationship on COURSE, one course acts as the *main course* and another as the *prerequisite course*.

Q27. Define Simple and Composite attributes.

- **Simple Attribute:** An attribute that cannot be divided into smaller sub-parts. Example: Age, Salary, Roll_No.
 - **Composite Attribute:** An attribute that can be divided into smaller sub-parts with independent meanings. Example: Name (divided into First_Name, Middle_Name, Last_Name) or Address (Street, City, State, Pincode).
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Q28. Define Single-valued and Multivalued attributes.

- **Single-valued Attribute:** An attribute that holds a single value for a given entity. Example: Date_of_Birth, SSN.
 - **Multivalued Attribute:** An attribute that can hold multiple values for a single entity. Represented by a double ellipse in ER diagrams. Example: Phone_Number (a person can have multiple numbers), College_Degrees.
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Q29. What is a Derived attribute? Give an example.

A **Derived Attribute** is an attribute whose value is not physically stored in the database but is computed dynamically from another attribute or data source. Represented by a dashed ellipse.

Example: Age (derived from Date_of_Birth and current date), Total_Salary (derived from Basic_Pay + HRA + DA).

Q30. What is a null value in the context of attributes?

A **NULL value** represents a special value used when an attribute value is **unknown**, **missing**, or **not applicable** for a particular entity.

Example: A student who does not own a mobile phone has a NULL value in the Phone_Number field.

Q31. Define Primary Key.

A **Primary Key** is a candidate key chosen by the database designer to uniquely identify each tuple (row) in a relation or entity in an entity set. It must contain unique values and cannot contain NULL values.

Q32. What is a strong entity set? What is a weak entity set?

- **Strong Entity Set:** An entity set that has a primary key sufficient to uniquely identify all its entities. Example: EMPLOYEE with primary key Emp_ID.
 - **Weak Entity Set:** An entity set that does not have a primary key of its own and depends on a strong (owner) entity set for its existence. Example: DEPENDENT (child/spouse of an employee).
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Q33. Define identifying entity/owner entity and identifying relationship.

- **Identifying/Owner Entity Set:** The strong entity set on which a weak entity set depends for existence and identification. Example: EMPLOYEE is the owner of DEPENDENT.
 - **Identifying Relationship:** The relationship set connecting a weak entity set to its identifying owner entity set. Represented by a double diamond. Example: HAS_DEPENDENT.
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Q34. What is a discriminator (partial key)?

A **Discriminator** (or **Partial Key**) is the set of attributes of a weak entity set that distinguishes among entities belonging to the same owner entity. Represented by a dashed underline.

Example: Dependent_Name in DEPENDENT — unique only for a specific employee's dependents.

Q35. Define Mapping Cardinality. What are its four types?

Mapping Cardinality (or Cardinality Ratio) expresses the number of entities in one entity set that can be associated with entities in another entity set via a relationship set.

Four Types:

1. One-to-One (1:1)
 2. One-to-Many (1:N)
 3. Many-to-One (N:1)
 4. Many-to-Many (M:N)
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Q36. What is Total Participation? What is Partial Participation?

- **Total Participation:** Every entity in the entity set MUST participate in at least one relationship in the relationship set (existence dependency). Represented by a double line in ER diagrams. Example: Every LOAN must belong to a CUSTOMER.
 - **Partial Participation:** Some entities in the entity set MAY NOT participate in the relationship set. Represented by a single line. Example: Not every EMPLOYEE manages a DEPARTMENT.
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Q37. What symbols represent entity sets, attributes, and relationship sets in ER diagrams?

- **Entity Set:** Rectangle
- **Weak Entity Set:** Double Rectangle

- **Attribute:** Ellipse / Oval
 - **Multivalued Attribute:** Double Ellipse
 - **Derived Attribute:** Dashed Ellipse
 - **Key Attribute:** Ellipse with underlined text
 - **Relationship Set:** Diamond
 - **Identifying Relationship:** Double Diamond
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Q38. Define Specialization and Generalization.

- **Specialization:** A top-down design process of designating sub-groupings (subclasses) within an entity set (superclass) based on specific distinguishing characteristics. Example: EMPLOYEE specialized into SALARIED_EMP and HOURLY_EMP.
 - **Generalization:** A bottom-up design process of combining multiple entity sets sharing common features into a single higher-level entity set (superclass). Example: Combining CAR and TRUCK into VEHICLE.
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Q39. What is Aggregation in the EER model?

Aggregation is an abstraction concept in the Enhanced ER (EER) model that allows a relationship set to be treated as a higher-level entity set so that it can participate in another relationship set. It solves the problem of expressing a relationship among relationships.

Q40. Define ISA relationship.

An **ISA ("Is A") relationship** represents superclass-subclass relationships in Specialization/Generalization hierarchies (e.g., SECRETARY *ISA* EMPLOYEE). It represents inheritance, where the subclass entity inherits all attributes and relationships of its superclass.

Q41. What is a redundant attribute? Give an example.

A **Redundant Attribute** is an attribute present in an entity set or relation that can be derived or obtained from other existing attributes or relationships, creating unnecessary duplication.

Example: Storing Dept_Name in STUDENT when Dept_ID is already stored and DEPARTMENT table maps Dept_ID to Dept_Name.

Q42. Define Key Constraint.

A **Key Constraint** specifies that an entity set or relationship set must have unique identifying values. In a relationship set, a key constraint on an entity set indicates that an entity can participate in at most one relationship instance (specifying 1:1 or 1:N cardinality).

Q43. What is an ER diagram? List its two main components.

An **Entity-Relationship (ER) Diagram** is a graphical representation of the logical structure of a database model.

Two Main Components:

1. **Entity Sets** (represented by rectangles)
 2. **Relationship Sets** (represented by diamonds linking entity sets)
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Q44. List the three components specified while mapping an ER model to a relational model.

1. **Relation Names** (tables derived from entity sets or relationship sets)
 2. **Attribute Names and Data Types** (columns of each table)
 3. **Primary Keys and Foreign Keys** (integrity constraints connecting tables)
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Q45. What is the primary key of a relation formed from a weak entity set?

The primary key of a relation formed from a weak entity set is the **combination of the primary key of its identifying owner entity set** (as a foreign key) **AND the discriminator (partial key)** of the weak entity set.

Example: Primary Key of DEPENDENT = (Emp_ID, Dependent_Name).

Q46. Name the four options available for mapping specialization/generalization to relations.

1. **Multiple relations — Superclass and Subclasses** (separate table for superclass, separate table for each subclass containing primary key + specific attributes).
 2. **Multiple relations — Subclasses only** (no table for superclass; each subclass table contains all inherited + specific attributes).
 3. **Single relation with one type attribute** (one unified table with a discriminator/type column and NULLs for inactive subclass attributes).
 4. **Single relation with multiple type attributes** (one unified table with boolean flag columns for overlapping subclasses).
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